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$$I = \int \frac{5^x}{5^x + 2} dx$$

$$\text{Stel: } t = 5^x \Rightarrow dt = 5^x \ln 5 dx \Leftrightarrow 5^x dx = \frac{dt}{\ln 5}$$

$$\begin{aligned} &= \int \frac{1}{5^x + 2} \cdot (5^x dx) = \int \frac{1}{t} \frac{dt}{\ln 5} \\ &= \frac{1}{\ln 5} \ln |t| + c = \frac{\ln |5^x + 2|}{\ln 5} + c \end{aligned}$$

References